

BRAC University
CSE341 : Microprocessors
Quiz#2

Duration : 25 Minutes

Total Marks : 20

ID:

Name:

Sec:

Consider the following assembly language code where DS: 1000,

MOV BX, [XXXX] where XXXX is the last 4 digits of your student ID. An Intel 8086 microprocessor takes yyyy ns(nanoseconds) to fetch, decode and execute this instruction, where yyyy is the first 4 digits of your student ID. Deduce the following from the given information:

- i. Length of 1 instruction cycle [2]
- ii. Length of 1 bus cycle [2]
- iii. The frequency at which 8086 is operating [4]
- iv. The time for T_{ON} and T_{OFF} [2]

Now, for the given instruction, deduce the value of A_0 and BHE'. Illustrate the timing diagram for the aforementioned operation (in question A) showcasing the AD0 - AD19, ALE, RD'/WR' (use either read or write) , Ready , and M/IO' (use either memory or I/O) pins. [2 + 8]